Markov Chains

Note Title 7/17/2017

- A collection of random variables $X = \{X_0, X_1, \ldots, \}$ is called a random process or a stochastic process.

- Suppose that, for each k, X_k takes values in a discrete set S. Unless otherwise stated, we will further assume that S is a finite set.

- X is called a Markov chain if

$$P(X_k = j \mid X_{k-1} = i, X_{k-2} = i_{k-2}, \ldots, X_0 = i_0)$$
$$= P(X_k = j \mid X_{k-1} = i) \quad \forall\, k, i, j, i_{k-2}, \ldots, i_0$$

- If $p_{ij} \triangleq P(X_k = j \mid X_{k-1} = i)$ does not depend on k, then X is

called a time-homogeneous MC. We will assume X is time-homogeneous unless otherwise stated.

- Let $p(k)$ be a row vector with its ith component being $p_i(k) = P(X_k = i)$. Clearly $\sum_{i \in S} p_i(k) = 1 \quad \forall k$.

- The MC is completely characterized by $p(0)$ and P, where P is called the probability transition matrix whose (i,j) element is P_{ij} which was defined earlier.

- Note that

$$p_j(k) = \sum_i P_{ij} p_i(k-1).$$

In matrix-vector form,

$$p(k+1) = p(k)\, P,$$

with the initial condition $p(0)$

Thus,

$$p(k) = p(0)\, P^{k}$$

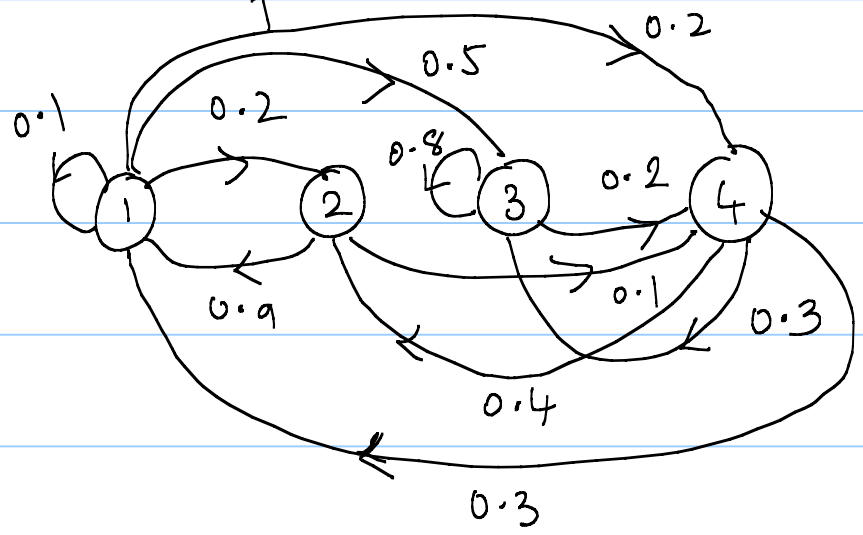
Example: $S = \{1, 2, 3, 4\}$

$$P = \begin{pmatrix} 0.1 & 0.2 & 0.5 & 0.2 \\ 0.9 & 0 & 0 & 0.1 \\ 0 & 0 & 0.8 & 0.2 \\ 0.3 & 0.4 & 0.3 & 0 \end{pmatrix}$$

Note that the row sum of each row of P is equal to 1 since

$$P_{ij} = P(X_k = j \mid X_{k-1} = i)$$

A MC is often represented by a transition diagram, which for this example is as follows:



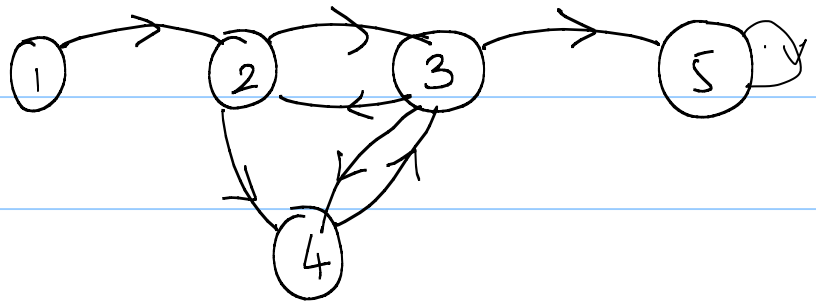
Thus, the transition diagram is a directed graph with an arrow wherever $P_{ij} > 0$.

□

- A state j is said to be reachable from a state i if there is a directed path from i to j.

- A set of states $\mathcal{C}$ is said to form a communicating class if $\forall i, j \in \mathcal{C}$, i is reachable from j, and it is a maximal such set, i.e., adding one state to $\mathcal{C}$ would mean that not all states in $\mathcal{C}$ are reachable from each other.

Example:



- $\{2, 3, 4\}$ form a communicating class.

  $\{5\}$ forms a communicating class.

  $\{1\}$ forms a communicating class if we define a state to be reachable from itself. □

- In the above example, state 1 is called

a transient state since there is a state j s.t. there is a non-zero probability of going from 1 to j but there prob. of going from j to 1 is zero. (here $j = 2, 3$ or 4)

- On the other hand, state 5 is called an absorbing state since you can never leave it.

- If a MC has a single communicating class, it is said to be irreducible.

Theorem: A finite-state irreducible MC has a unique stationary distribution, i.e., $\exists$ a probability row vector μ s.t.

$$\mu = \mu P.$$

Example:



$P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$; not irreducible.

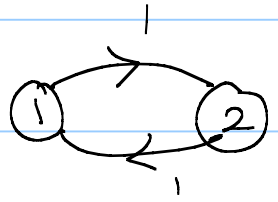
Therefore, does not have a unique stationary distribution since every probability vector μ satisfies $\mu = \mu P$.

□

Suppose that a MC is irreducible, does $\lim_{k \to \infty} p(k) = \mu$?

Example:

$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$



Suppose $p(0) = (1 \quad 0)$, i.e., the MC starts in state 1. Then

$$p(k) = \begin{cases} (1 \quad 0) & \text{if } k \text{ is even} \\ (0 \quad 1) & \text{if } k \text{ is odd.} \end{cases}$$

□

- We say that it is possible to return to state i in $n \geq 1$ steps if

$$P(X_n = i \mid X_0 = i) > 0$$

- The period d_i of a state i is defined to be

$$d_i = \gcd\{ n : p_{ii}^{(n)} > 0 \} \text{ where}$$

$$p_{ii}^{(n)} = P(X_n = i \mid X_0 = i)$$

- State is said to be aperiodic if its period is one.

- If one state of an irreducible MC is aperiodic, then all states are aperiodic. In this case, the MC is

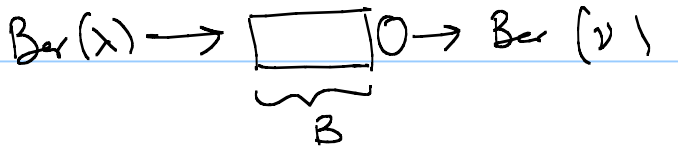
said to be aperiodic.

Theorem: For an irreducible, aperiodic finite-state MC,

$$\lim_{k \to \infty} p(k) = \mu \qquad \forall\, p(0).$$

□

Example: Consider a queue with Ber(λ) arrivals and Ber(ν) service. Assume that the queue can hold up to B packets, and assume that arrivals occur before departures in each time slot.



X_k: # packets in the buffer at the beginning of time slot k (before any arrivals)

Suppose $X_k = i$, where $i \neq 0$ or B.

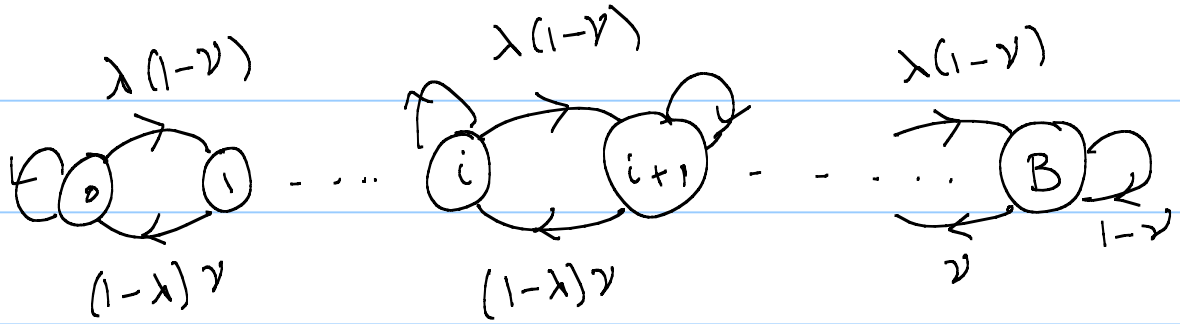
$$X_{k+1} = \begin{cases} X_k & \text{w.p. } (1-\lambda)(1-\nu) + \lambda\nu \\ & \text{(no arrivals \& departures or one arrival \& one departure)} \\ X_k + 1 & \text{w.p. } \lambda(1-\nu) \\ X_k - 1 & \text{w.p. } (1-\lambda)\nu \end{cases}$$

Suppose $X_k = 0$.

$$X_{k+1} = \begin{cases} 0 & \text{w.p. } (1-\lambda) + \lambda\nu \\ 1 & \text{w.p. } \lambda(1-\nu) \end{cases}$$

Suppose $X_k = B$,

$$X_{k+1} = \begin{cases} B-1 & \text{w.p. } \nu \\ B & \text{w.p. } 1-\nu \end{cases}$$



If we assume $\lambda, \nu \in (0,1)$, then this is an irreducible MC since there is a path from any i to any j in the above directed graph.

Further, since the MC can stay in any state w.p. >0, it is also aperiodic $\Rightarrow$ the MC has a stationary μ which is reached in steady-state.

This MC has the property

$$P_{ij} = 0 \quad \text{unless} \quad |i-j| \leq 1.$$

Such MCs are called skip-free MCs and this allows for computation of μ. Note that $\mu = \mu P \Rightarrow$

$$\mu_0 = \mu_0 P_{00} + \mu_1 P_{10}$$

$$\Leftrightarrow \quad \mu_0 (1 - P_{00}) = \mu_1 P_{10}$$

$$\Leftrightarrow \quad \mu_0 P_{01} = \mu_1 P_{10} \qquad \text{— (A)}$$

$$\mu_1 = \mu_0 P_{1} + \mu_1 P_{11} + \mu_2 P_{21}$$

$$\Leftrightarrow \quad \mu_1 (1 - P_{11}) = \mu_0 P_{01} + \mu_2 P_{21}$$

$$\Leftrightarrow \quad \mu_1 (P_{10} + P_{12}) = \mu_0 P_{01} + \mu_2 P_{21}$$

from (A)

$$\Leftrightarrow \quad \mu_1 P_{12} = \mu_2 P_{21}$$

Continuing as above, we get

$$\mu_i P_{i,i+1} = \mu_{i+1} P_{i+1,i}$$

$$\forall\, i \leq B-1$$

$$\Rightarrow \quad \mu_1 = \frac{P_{01}}{P_{10}} \mu_0$$

$$\mu_2 = \frac{P_{12}\, \mu_1}{P_{21}}$$

$$\mu_B = \frac{p_{B-1,B}\,\mu_{B-1}}{p_{B,B-1}}$$

This along with $\mu_0 + \cdots + \mu_B = 1$ gives us enough equations to solve for $\pi_0, \ldots, \pi_B$ (in this example, one can even get a closed-form expression).

□

Example: Consider two queues which provide $\text{Ber}(\nu_1)$, $\text{Ber}(\nu_2)$ service, respectively. Assume that arrivals occur according to $\text{Ber}(\lambda)$ and an arrivals join the shorter of the two queues, with ties broken at random. As before, assume each queue has a max size of B and arrivals occur before departures in each

time slot. This is a more complicated MC with X_k = (length of queue 1, length of queue 2)

Suppose $X_k = (3, 2)$ and suppose $B = 10$.

Then

$$X_{k+1} = \begin{cases} (3,3) & \text{w.p. } \lambda(1-\nu_1)(1-\nu_2) \\ (3,2) & \text{w.p. } \lambda(1-\nu_1)\nu_2 + (1-\lambda)(1-\nu_1)(1-\nu_2) \\ (2,3) & \text{w.p. } \lambda\nu_1(1-\nu_2) \\ (2,2) & \text{w.p. } \lambda\nu_1\nu_2 \\ (2,1) & (1-\lambda)\nu_1\nu_2 \\ (3,1) & (1-\lambda)(1-\nu_1)\nu_2 \\ (2,2) & (1-\lambda)\nu_1(1-\nu_2) \end{cases}$$

Similar transitions will happen if $X_k = (i, j)$ where neither i nor j

belong to $\{0, B\}$. The boundary cases need to be handled carefully.

Equivalent definition of periodicity | aperiodicity

$d_i > 1$ is said to be the period of state i if $p_{ii}^{(n)} = 0$ when n is not a multiple of d_i and d_i is the largest such number. If no such d_i exists, then i is aperiodic.

Here, $p_{ii}^{(n)} = \text{Prob}\left(X_n = i \mid X_0 = i \right)$